

```
static int test_init(void) {
    struct task_struct *task; #1
    for_each_process(task) { #2
        printk("Process Name: %s\t PID:%d\t Process State:%ld\n",
            task->comm, task->pid, task->state); #3
    }

    return 0;
}

static void test_exit(void) {
    printk(KERN_INFO "Cleaning Up.\n");
}

module_init(test_init);
module_exit(test_exit);
```

Explanations:

1. A pointer named *task* was created to the structure *task_struct*.
2. You iterate over the processes handled by the kernel (whether they are in executing or waiting state).
3. You access the structure through the pointer and display

its properties.

Make the necessary changes in the old *Makefile*, compile, and insert it. You should see something like what's shown in Figure 3. You will see process names along with their process IDs and the process state (integer value; and Google for what those values mean). There are over 50 properties associated with processes defined in that structure. So take a reading at the beautifully documented structure and start playing with processes.

Linux kernel module programming is the best thing I have come across so far. I am learning it in depth, and will share my experiences in the future too. A large number of e-books are also available online. The power of Google is tremendous. So explore them. Queries and suggestions are always welcome. **END** 

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The author loves to explore open source technologies and enjoys programming. He has learned Python, PHP, C and C++. He loves rock and metal music, and plays the guitar too. He runs a blog related to Linux at www.fossstuff.wordpress.com and can be contacted via ankur.aggarwal2390@gmail.com or on Twitter as @ankurtwi.

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